6b. Course Wrap-Up

PhD Applied Methods

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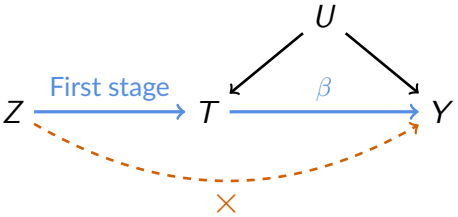
The IV setup

We want the causal effect of treatment T on outcome Y, but there is a confounder U.

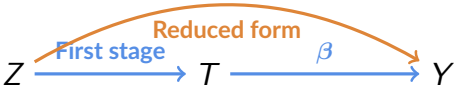
An **instrument** Z provides exogenous variation in T:



Z affects Y **only through** T	(exclusion restriction)
Z is unrelated to U	(exogeneity)

The Wald ratio: reduced form ÷ first stage

The DAG tells us that Z's effect on Y travels **entirely** through T:



Chain rule:

$$\underbrace{\frac{\partial Y}{\partial Z}}_{\text{Reduced form}} = \underbrace{\frac{\partial T}{\partial Z}}_{\text{First stage}} \times \underbrace{\beta}_{\text{Causal effect}}$$

Rearranging:

$$\beta_{\text{Wald}} = \frac{\text{Reduced form}}{\text{First stage}} = \frac{\mathbb{E}[Y_i|Z_i=1] - \mathbb{E}[Y_i|Z_i=0]}{\mathbb{E}[T_i|Z_i=1] - \mathbb{E}[T_i|Z_i=0]}$$

But who does β_{Wald} apply to?

Not everyone responds to the instrument the same way.

Think of a **scholarship** (Z) designed to push people into **college** (T):

Group	$T_i(Z{=}0)$	$T_i(Z{=}1)$	**Behavior**
Compliers	0	1	Go to college *because of* scholarship
Always-takers	1	1	Would go regardless
Never-takers	0	0	Won't go regardless
Defiers	1	0	Scholarship *deters* them (?!)

The instrument only **changes behavior** for compliers (and defiers). Always-takers and never-takers contribute nothing to the first stage.

Decomposing the reduced form by group

The reduced form can be written as:

$$\underbrace{\mathbb{E}[Y|Z{=}1] - \mathbb{E}[Y|Z{=}0]}_{\text{Reduced form}} = \mathbb{E}\big[(T_i(1) - T_i(0))(Y_i(1) - Y_i(0))\big]$$

Each group contributes according to how Z changes their treatment (ΔT):

$$\begin{aligned}
&= \underbrace{(+1)}_{\Delta T} \cdot \mathbb{E}[Y_i(1) - Y_i(0)|\text{Complier}] \cdot P(\text{Complier}) \\
&+ \underbrace{(0)}_{\Delta T} \cdot \mathbb{E}[Y_i(1) - Y_i(0)|\text{Always/Never}] \cdot P(\text{Always/Never}) \\
&+ \underbrace{(-1)}_{\Delta T} \cdot \mathbb{E}[Y_i(1) - Y_i(0)|\text{Defier}] \cdot P(\text{Defier})
\end{aligned}$$

Why only compliers (and defiers) matter

Always-takers and never-takers **drop out**: their $\Delta T = 0$.

Only compliers and defiers generate variation in T in response to Z:

$$\begin{aligned}\text{Reduced form} = &(+1) \cdot \mathbb{E}[Y_i(1) - Y_i(0) | \text{Complier}] \cdot P(\text{C}) \\ &(-1) \cdot \mathbb{E}[Y_i(1) - Y_i(0) | \text{Defier}] \cdot P(\text{D})\end{aligned}$$

Notice: compliers get weight $+1$, defiers get weight -1.

This is a **non-convex** combination — the defier term enters with a negative sign!

Monotonicity eliminates defiers

Monotonicity assumption: $T_i(1) \geq T_i(0)$ for all i

Everyone weakly responds to the instrument in the same direction. $\implies$ No defiers.

Then the reduced form becomes:

$$\mathbb{E}[Y|Z{=}1] - \mathbb{E}[Y|Z{=}0] = \mathbb{E}[Y_i(1) - Y_i(0)|\text{Complier}] \cdot P(\text{Complier})$$

And the first stage is simply:

$$\mathbb{E}[T|Z{=}1] - \mathbb{E}[T|Z{=}0] = P(\text{Complier})$$

The LATE result

Dividing the reduced form by the first stage:

$$\beta_{\text{Wald}} = \frac{\text{Reduced form}}{\text{First stage}} = \mathbb{E}[Y_i(1) - Y_i(0) | \text{Complier}] = \textbf{LATE}$$

Local Average Treatment Effect: the average causal effect among **compliers** — those who change treatment status in response to the instrument.

Key implications:

- LATE $\neq$ ATE in general (unless effects are homogeneous)
- Different instruments identify different complier populations $\rightarrow$ different LATEs
- We cannot name the compliers — only estimate their share from the first stage

Outline

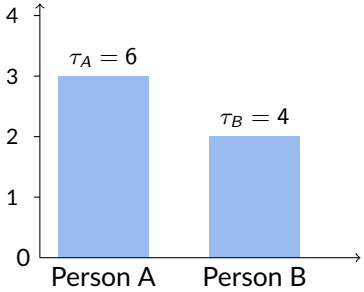
1. Recap: Instrumental Variables and the Wald Ratio

2. Why Negative Weights Break Identification

3. Course Recap

A simple weighted average

Suppose two people each have a **positive** treatment effect:



With **positive weights** that sum to 1:

$$\bar{\tau} = w_A \cdot \tau_A + w_B \cdot \tau_B$$

For example, $w_A = 0.5$, $w_B = 0.5$:

$$\bar{\tau} = 0.5 \times 6 + 0.5 \times 4 = 5$$

For **any** $w_A, w_B \geq 0$ with $w_A + w_B = 1$:

$$4 \leq \bar{\tau} \leq 6$$

The average is always **between** the individual effects.

What if a weight is negative?

Now suppose $w_A = 1.5$ and $w_B = -0.5$:

$$\bar{\tau} = 1.5 \times 6 + (-0.5) \times 4 = 7$$

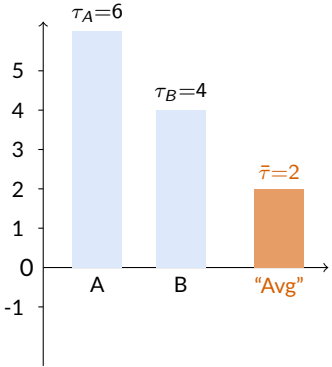
The "average" is **outside** $[4, 6]$.

Even worse: $w_A = -0.5$ and $w_B = 1.5$:

$$\bar{\tau} = (-0.5) \times 6 + 1.5 \times 4 = 3$$

Or $w_A = -1$ and $w_B = 2$:

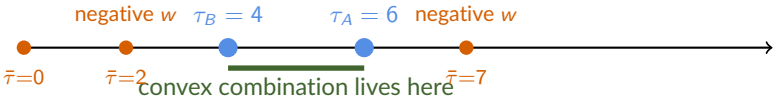
$$\bar{\tau} = (-1) \times 6 + 2 \times 4 = 2$$



With negative weights, the "average" can be **anywhere** — even negative when all individual effects are positive!

Convex combinations guarantee sensible averages

A convex combination requires all weights $w_i \geq 0$ and $\sum_i w_i = 1$.
This guarantees the weighted average lies **within the range** of individual values.



If every individual treatment effect is positive, a convex combination **must be positive**.

With negative weights, you can get **zero or negative** "averages" — meaningless!

Where negative weights appear in this course

This issue shows up in **two** major contexts we studied:

1. Instrumental variables — defiers

- Compliers contribute weight $+P(\text{C})$, defiers contribute weight $-P(\text{D})$
- Without monotonicity (no defiers), the Wald ratio mixes complier and defier effects with **opposite signs**
- $\implies$ Cannot interpret the IV estimate as *any* group's treatment effect

2. Difference-in-differences — staggered timing with TWFE

- TWFE puts **negative weights** on some cohort-time effects when it uses already-treated units as controls
- $\hat{\beta}^{TWFE} = \sum_{g,t} w_{g,t} \cdot ATT_{g,t}$ where some $w_{g,t} < 0$
- $\implies$ Can estimate a **negative** overall effect even when all $ATT_{g,t} > 0$

The fix in both cases: ensure your estimator uses a **convex combination** of causal effects.

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Lecture 1: Potential Outcomes and Randomization

Core framework: The potential outcomes model (Rubin causal model)

- Each unit has two potential outcomes: $Y_i(1)$ and $Y_i(0)$
- **Fundamental problem**: We only observe one — the counterfactual is missing
- Naïve comparisons conflate treatment effects with **selection bias**

Solution: Randomized controlled trials (RCTs)

- Random assignment $\implies \mathbb{E}[Y(0)|D{=}1] = \mathbb{E}[Y(0)|D{=}0]$
- Simple difference in means identifies the ATE
- Practical issues: balance testing, stratification, clustered randomization, SUTVA, power calculations

When compliance is imperfect: ITT and the Wald estimator preview IV

Lecture 2: Instrumental Variables

Problem: Treatment is endogenous — selection on unobservables

- Find a variable Z that affects T but has no direct effect on Y (exclusion restriction)
- Wald estimator: $\beta =$ Reduced form $\div$ First stage
- Generalization: Two-Stage Least Squares (2SLS)

Key challenges:

- Exclusion restriction is **untestable** — must be argued, not verified
- With heterogeneous effects: IV estimates the **LATE**, not the ATE
- Weak instruments bias IV toward OLS — check first-stage F-statistic

Takeaway: IV is powerful but requires strong assumptions. Always think carefully about who the compliers are and whether the exclusion restriction is credible.

Lecture 3: Difference-in-Differences

Setting: Panel data with treatment that turns on at a specific time

- Compare changes over time between treated and control groups
- **Parallel trends**: In the absence of treatment, both groups would have followed the same trajectory
- Basic 2×2 DiD; regression with TWFE (unit + time fixed effects)

Modern developments:

- Event study specifications to examine dynamic effects and test pre-trends
- **Staggered timing problem**: TWFE can give negative weights with heterogeneous effects
- Solutions: Callaway & Sant'Anna, de Chaisemartin & D'Haultfoeuille, Sun & Abraham
- Synthetic DiD: combine synthetic control with DiD for better counterfactuals

Lecture 4: Regression Discontinuity Design

Setting: Treatment assigned by a cutoff on a running variable Z_i

- **Sharp RDD**: $D_i = \mathbb{1}\{Z_i \geq z^*\}$ — compare outcomes just above vs. just below
- Identification via **continuity**: $\mathbb{E}[Y(0)|Z{=}z]$ smooth at z^*
- Local linear regression; bandwidth selection (bias-variance trade-off)

Fuzzy RDD: Treatment probability jumps at cutoff but not from 0 to 1

- Ratio of jumps = LATE at the cutoff (same logic as Wald/IV)

Validity checks:

- McCrary density test (no manipulation of running variable)
- Covariate balance at the cutoff
- Results robust to bandwidth choice

Lecture 5: Empirical Tools

Practical issues that arise across all causal inference methods:

- **Clustering**: Standard errors when observations are correlated within groups
- **Power**: Sample size calculations and minimum detectable effects
- **Attrition**: When units drop out of the sample; Lee bounds
- **Multiple testing**: Family-wise error rate corrections
- **Robustness**: Sensitivity analysis; replication
- **Spillovers**: When treatment affects control units (SUTVA violations)
- **Measurement**: Classical and non-classical measurement error

The toolkit: matching methods to settings

Method	Key assumption	When to use
RCT	Random assignment	You control the experiment
IV / 2SLS	Exclusion restriction	Exogenous variation in T
DiD	Parallel trends	Panel data; staggered treatment
RDD	Continuity at cutoff	Treatment assigned by threshold

Common thread: Every method constructs a credible **counterfactual** for the treated group.

The differences are in **what you assume** to build that counterfactual, and **what population** the estimated effect applies to.

Final thoughts

- **No method is assumption-free**: The key question is always whether the assumptions are credible in your setting
- **Design matters more than estimation**: The most sophisticated estimator cannot fix a flawed research design
- **Transparency**: Show your readers why your identifying assumptions are plausible — pre-trends, balance tests, placebo checks, robustness to specification
- **Know your estimand**: Are you estimating the ATE, ATT, or LATE? For which subpopulation? This shapes both the method you choose and how you interpret results